
DP-900: Microsoft Data Fundamentals

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Core data concepts and related Microsoft Azure data services

If you're a data analyst, junior data engineer, database administrator, or app developer, this exam will demonstrate your ability to navigate Azure's features. Furthermore, if you need to demonstrate your proficiency with Microsoft data management and visualization products, the DP-900 is an excellent place to start.

Extreme labs – and cert

- [Exam DP-900: Microsoft Azure Data Fundamentals - Certifications | Microsoft Learn](#)
- [Course DP-900T00--A: Microsoft Azure Data Fundamentals - Training | Microsoft Learn](#)
- [MicrosoftLearning/DP-900T00A-Azure-Data-Fundamentals: DP-900 ILT lab instructions \(github.com\)](#)
- [Study guide for Exam DP-900: Microsoft Azure Data Fundamentals | Microsoft Learn](#)
- DP 900 – Boot Camp - [Collections | Microsoft Learn](#)

Pre 12/19/2022 – much has changed with 2024 AI Services & Newer BUP Management 12/15/2024 – 02/22/2026

Overview

Data is the “oil of the 21st Century Digital Economy”

[An Overview of Microsoft Azure Services | Splunk](#)



Elasticity in this case is the ability to provide additional compute resource when needed and reduce the compute resource when not needed to reduce costs. Autoscaling is an example of elasticity.



Advisor



Machine Learning

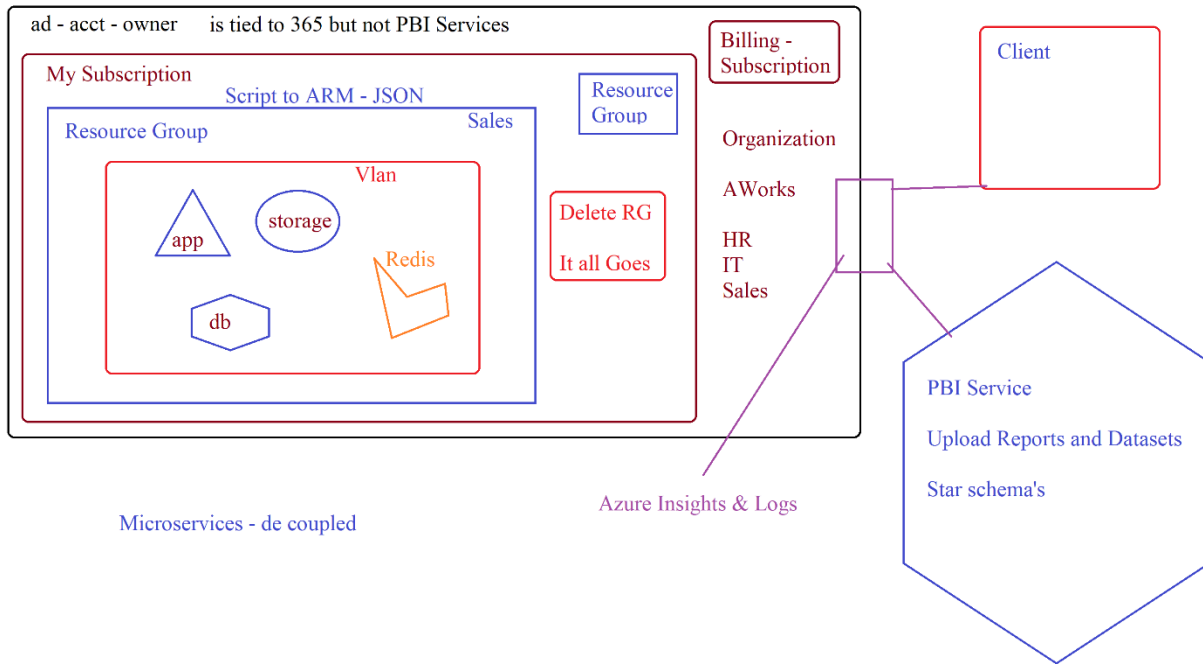


Cosmos DB



Azure DevOps

- Cloud applications make use of multiple Azure services.
- There is **no single path** to learn these services independently.



What is:

1. PaaS
 - a. Azure App Service is a platform-as-a-service (PaaS) offering that lets you create web and mobile apps for any platform or device and connect to data anywhere
2. IaaS
3. SaaS
4. DaaS

Concepts

Edge to cloud: Place computing resources closer to the user or the device, at the “edge” of the network, rather than in a hyperscale cloud data center that might be many miles away in the “core” of the network.

Microservices are a software approach that creates applications as a loose coupling of specific services or functions, rather than as a single, “monolithic” program.

Availability and fault tolerance with regions and zones

Messaging - stream processing architecture

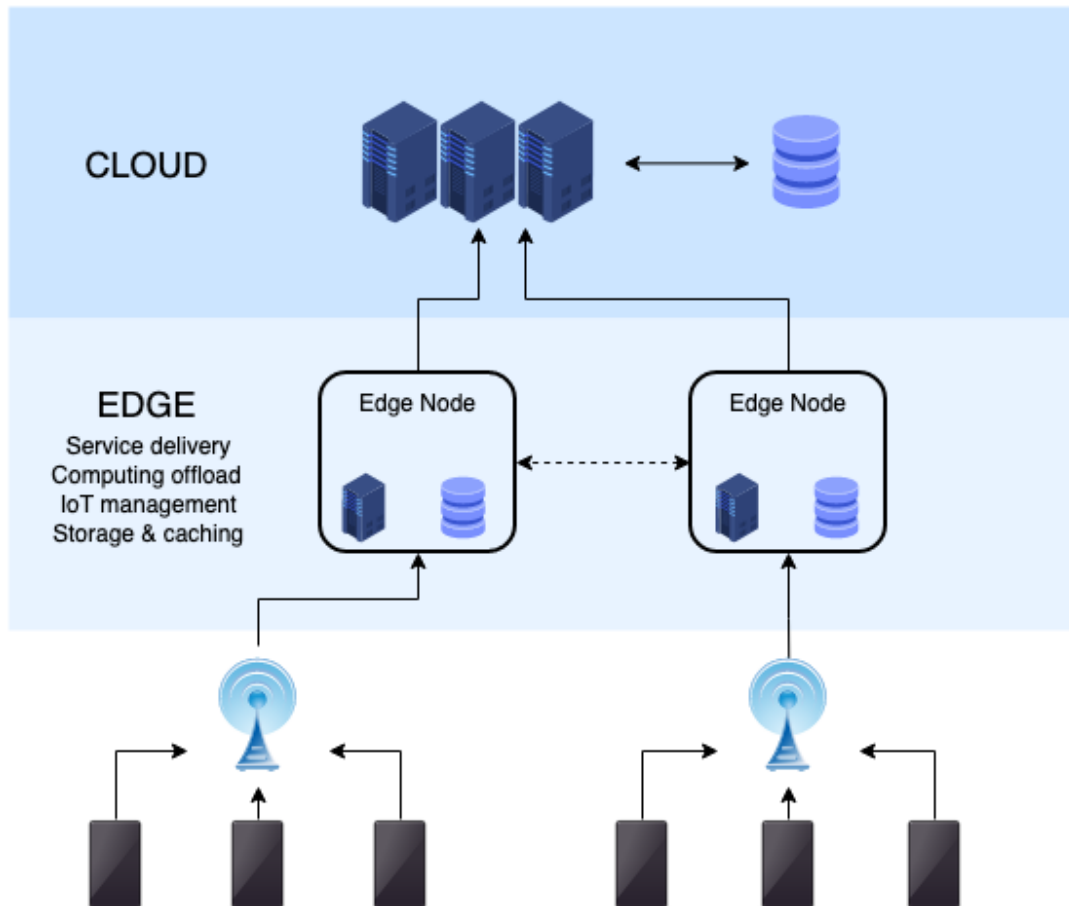
Azure application services - develop, deploy, maintain:

- Azure AI
- Azure Analytics
- Azure IoT
- Azure Active Directory (AD)

- Azure Media Services
- Azure Scheduler

Azure IoT: In Edge-to-cloud technologies, or the “Internet of Things,” a “thing” could be from a coffee pot to a car, as long as it can connect and communicate over the internet with other devices without human intervention.

- Azure’s IoT Central (SaaS) offers pre-configured solutions.
- Azure IoT solution accelerators offer more flexibility to develop a custom solution for your organization.
- Azure IoT Hub is the core platform (PaaS) supporting Azure IoT Central and Azure IoT solution accelerators. (zero-touch device provisioning)



Azure data services

[store and manage data](#) on the cloud. Microsoft Azure comes with a range of data services:

- Azure Storage
- Azure SQL Database
- Azure DocumentDB
- Azure StorSimple
- Azure Redis Cache

Azure DevOps automates your software delivery process and boosts your software development by providing an alternative to the self-managed CI/CD servers and open-source DevOps tools.

Azure compute services are the hosting services responsible for hosting and running the application workloads. These include Azure Virtual Machines (VMs), Azure Container Service, Azure App Services, Azure Batch and Azure ServiceFabric.

Azure Virtual Machine (VM) is an on-demand, scalable computing resource.

Azure Container Service

A container is a system allowing software to be made modular, portable and standardized so it can be easily deployed on any computing environment. Containers are designed to contain an application's code as well as all of its dependencies so that everything needed to run the application is in a single place. What is a Container? A Beginner's Guide | Splunk

Azure helps you leverage the container-based development practices and the microservices architecture. You can migrate your .NET applications to microservices using Windows Server containers with Azure Service Fabric. Further, you can use Azure Kubernetes Service to scale and orchestrate Linux containers.

You can choose between Docker Hub and Azure Container Registry to store your images and deploy to any preferred target. Moreover, it simplifies the configuration process and optimizes it for the cloud. The major advantage is that it consumes less space as compared to VMs and starts instantly; hence speeding up the processes.

Azure network services

Azure Network Services refer to those services that perform networking operations within Azure and between Azure and on-premises infrastructure. These include Azure Virtual Network, Azure ExpressRoute, Azure-provided DNS, Azure Traffic Manager and Azure Content Delivery Network.

Azure Content Delivery Network (CDN)

Azure CDN allows distribution of large files by providing high bandwidth content and reducing the file download time. It caches the file to an Azure data center closest to the user's geographical location. Hence, it lowers the latency to a great extent and decreases the load on application by relieving it from the whole process of delivering the content. The end-users derive the main benefit out of this service as it offers a better experience to them.

Azure Cosmos DB

Azure Cosmos DB is a global-scale non-relational (NoSQL) database system that supports multiple application programming interfaces (APIs), enabling you to store and manage data as JSON documents, key-value pairs, column-families, and graphs.

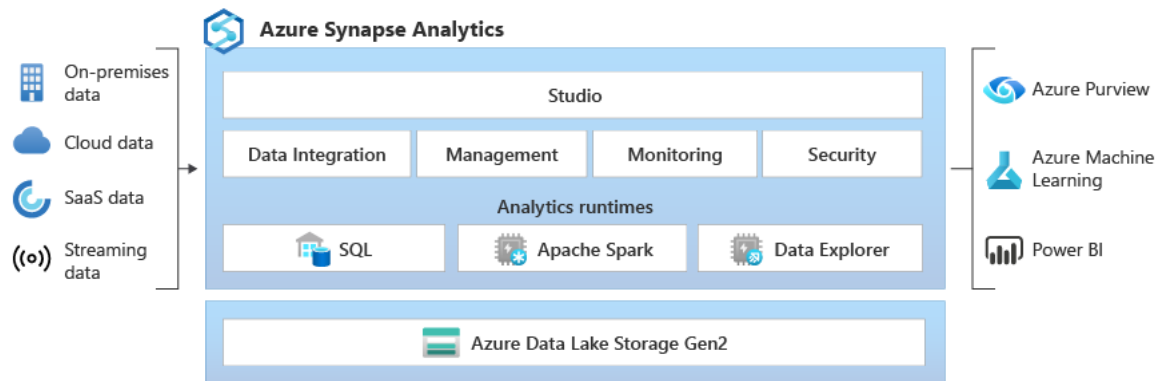
Azure Data Factory

Azure Data Factory logo Azure Data Factory is an Azure service that enables you to define and schedule data pipelines to transfer and transform data.

Azure Synapse Analytics

Azure Synapse Analytics logo Azure Synapse Analytics is a comprehensive, unified data analytics solution that provides a single service interface for multiple analytical capabilities.

[Tutorial: Get started with Azure Synapse Analytics - Azure Synapse Analytics | Microsoft Learn](#)



Azure Databricks

Azure Databricks logo Azure Databricks is an Azure-integrated version of the popular Databricks platform, which combines the Apache Spark data processing platform with SQL database semantics and an integrated management interface to enable large-scale data analytics.

Azure HDInsight

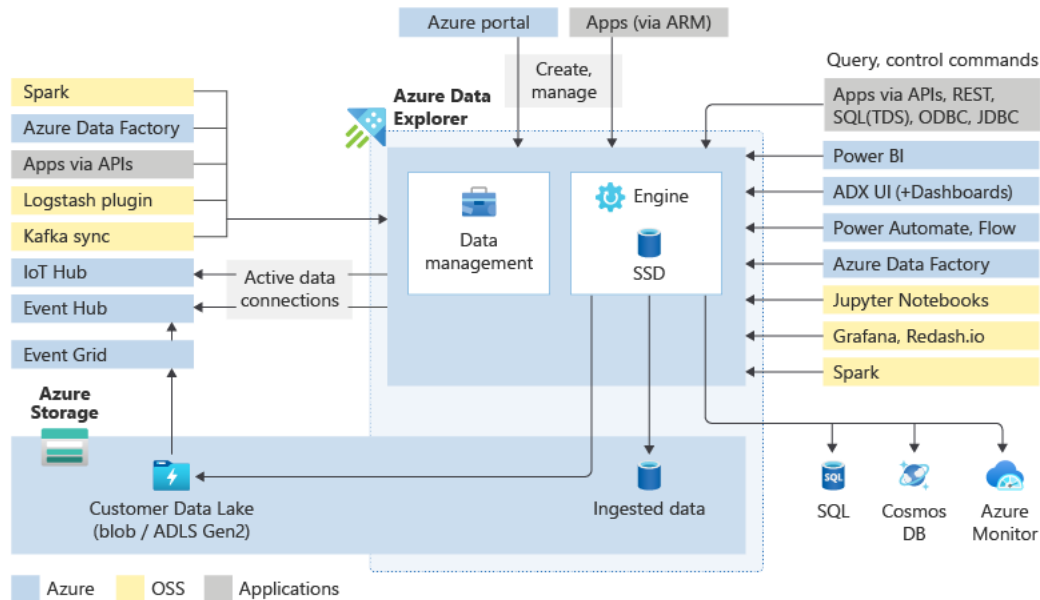
Azure HDInsight logo Azure HDInsight is an Azure service that provides Azure-hosted clusters for popular Apache open-source big data processing technologies.

Azure Stream Analytics

Azure Stream Analytics logo Azure Stream Analytics is a real-time stream processing engine that captures a stream of data from an input, applies a query to extract and manipulate data from the input stream, and writes the results to an output for analysis or further processing.

Azure Data Explorer

Azure Data Explorer logo Azure Data Explorer is a standalone service that offers the same high-performance querying of log and telemetry data as the Azure Synapse Data Explorer runtime in Azure Synapse Analytics. Query and analyze data that includes a timestamp attribute



Microsoft Purview

Azure Purview logo Microsoft Purview provides a solution for enterprise-wide data governance and discoverability. You can use Microsoft Purview to create a map of your data and track data lineage across multiple data sources and systems, enabling you to find trustworthy data for analysis and reporting. Enforce data governance.

Microsoft Power BI

Microsoft Power BI logo Microsoft Power BI is a platform for analytical data modeling and reporting that data analysts can use to create and share interactive data visualizations.

✓ DP-900 Azure Data Fundamentals — Clean Workflow Task List

Purpose: A minimal, structured sequence of tasks that touches *every exam requirement* without drifting into overlapping / advanced Azure content.

1 Core Data Concepts (Start Here — Foundation)

Tasks

- Understand types of data
 - Structured
 - Semi-structured
 - Unstructured
- Understand transactional vs. analytical workloads
 - OLTP
 - OLAP
- Learn batch vs. streaming data
- Know basic data roles
 - Data engineer
 - DBA
 - Data analyst
 - Data architect

Avoid the rabbit hole: You don't need to learn database internals (indexes, ACID deep dives, etc.). DP-900 only requires conceptual definitions.

2 Relational Data on Azure

Tasks

- Know what a relational database is
- Understand tables, rows, columns, keys
- Learn common Azure relational services:
 - Azure SQL Database
 - Azure SQL Managed Instance

- SQL Server on Azure VM
- Understand basic T-SQL concepts (no writing):
 - SELECT, INSERT, UPDATE
 - Joins (conceptual)

Avoid the rabbit hole: Don't deep-dive T-SQL or Azure SQL architecture options — DP-900 only tests awareness.

3 Non-Relational (NoSQL) Data on Azure

Tasks

- Learn types of NoSQL databases
 - Key-value
 - Document
 - Column-family
 - Graph
- Know Azure Cosmos DB supports **all** of these
- Understand global distribution + multi-region replication (conceptual)
- Know throughput & consistency as concepts

Avoid the rabbit hole: No need to understand consistency models beyond “strong vs eventual.”

4 Analytics Workloads (Big Data Concepts)

Tasks

- Know what big data is
- Understand distributed processing
 - Spark (high level)
 - MPP / scale-out systems
- Recognize Azure analytics tools:
 - Azure Synapse Analytics
 - Azure Data Explorer

- Azure Databricks
- Azure HDInsight (legacy but still tested)

Avoid the rabbit hole: You don't need to understand Spark code, workspace setup, pipelines, or cluster types.

5 Data Ingestion + Transformation

Tasks

- Learn the purpose of ETL vs ELT
- Know ingestion tools:
 - Azure Data Factory
 - Synapse Pipelines
- Understand data integration concepts:
 - Pipelines
 - Activities
 - Data movement
 - Triggers (conceptual)

Avoid the rabbit hole: No need to build actual ADF pipelines or know mapping data flows.

6 Data Storage Options (Conceptual)

Tasks

- Know Azure Storage types:
 - Blob storage
 - Files
 - Tables (key-value)
 - Queues
- Understand hot / cool / archive
- Understand structured vs unstructured placement

Avoid the rabbit hole: Ignore storage tier pricing details — just know the purpose of each tier.

7 Real-Time Stream Processing

Tasks

- Know streaming scenarios (IoT, telemetry)
- Azure Stream Analytics (concept only)
- Event Hubs vs IoT Hub
 - Event Hubs → high-throughput ingestion
 - IoT Hub → device-centric bi-directional messaging

Avoid the rabbit hole: Don't dive into consumer groups or partitioning — high level only.

8 Data Governance & Security (Microsoft Purview)

Tasks

- Understand what data governance is
- Know Microsoft Purview provides:
 - Data catalog
 - Lineage
 - Classification
- Know basics of Azure security:
 - RBAC
 - Data encryption
 - Firewall concepts (very light)

Avoid the rabbit hole: Skip complex role structures and networking configuration topics.

9 Data Visualization (Power BI Basics)

Tasks

- Understand what Power BI is
- Know the differences:
 - Power BI Desktop

- Power BI Service
 - Understand datasets, reports, dashboards (definition only)

Avoid the rabbit hole: No need to build reports, learn DAX, or study licensing.

✓ DP-900 Billable Azure Services & Billing Units (Short Table)

Azure Service	What You Pay For	Billing Unit	Citation
Azure SQL Database	Compute + storage depending on pricing model	DTU (bundled compute/storage/IO) or vCore (compute per core)	[learn.microsoft.com]
Azure Cosmos DB	Compute (throughput), storage, bandwidth	Request Units per second (RU/s) for NoSQL APIs; vCores for PostgreSQL/MongoDB API	[azure.microsoft.com] , [azure.cn]
Azure Synapse Analytics	Serverless SQL: data scanned; Dedicated SQL: compute; Spark: vCore-hours; Pipelines: activity runs + IR hours	Per TB scanned (serverless), DWU-hour (dedicated), vCore-hour (Spark), Per activity run / IR hour (pipelines)	[azure.microsoft.com] , [techbrothersit.com]
Azure Data Factory	Pipeline orchestration, activity runs, integration runtime compute, data movement	Per activity run, Integration Runtime hour, Data Integration Unit (DIU)-hour	[azure.microsoft.com] , [atmosera.com]
Azure Storage (Blob)	Storage capacity + operations + egress	Per GB stored (Hot/Cool/Archive), per 10,000 operations, per-GB data retrieval (Cool/Cold/Archive)	[learn.microsoft.com] , [clouddaily.co]
Azure Stream Analytics	Provisioned compute for stream processing	Streaming Unit (SU) / SU-hour	[azure.microsoft.com]

References

1. [Cloud Computing Terms | Microsoft Azure](#)
2. [Notebooks at Microsoft - Visual Studio](#)
3. [What Is Edge Computing and Why Is It Important? | Microsoft Azure](#)
 - a. Allows devices in remote locations to process data at the "edge" of the network, either by the device or a local server.
 - b. [Edge Computing - Microsoft Research](#)
4. [Azure IoT – Internet of Things Platform | Microsoft Azure](#)
5. [What is Azure SQL Edge? | Microsoft Learn](#)
6. [SQL Server \(mssql\) - Visual Studio Marketplace](#) – extension for VS code
7. [Use the Visual Studio Code mssql extension - SQL Server | Microsoft Learn](#)
8. [Native scoring with T-SQL PREDICT - SQL machine learning | Microsoft Learn](#)
9. [Azure Synapse Analytics - Azure Synapse Analytics | Microsoft Learn](#)
10. [Introduction to Azure Data Explorer - Training | Microsoft Learn](#)
11. [Add your custom domain - Azure Active Directory - Microsoft Entra | Microsoft Learn](#)
12. [Write your first query with Kusto Query Language - Training | Microsoft Learn](#)
13. End List